



Numerical Analysis Project

Air Quality Prediction

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Air Pollution

1 Introduction

- The current criteria pollutants are Carbon Monoxide (CO), Lead (Pb), Nitrogen Dioxide (NO₂), Ozone (O₃), Particulate matter (PM), Sulfur Dioxide (SO₂)
- The effects of air pollution include short-term health effect, long-term health effect, environment effect



AQI

1 Introduction

Air quality index (AQI) which quantifies air quality in a region is used to express how polluted the air is currently or how polluted it is forecasted to become. There are five steps of AQI:

AIR QUALITY INDEX - PARTICULATE MATTER	
301+	HAZARDOUS
201-300	VERY UNHEALTHY
151-200	UNHEALTHY
101-150	UNHEALTHY FOR SENSITIVE GROUPS
51-100	MODERATE
0-50	GOOD

Figure: Air Quality index



Calculate AQI

1 Introduction

Domestic Air Quality Index (AQI) is calculated as follows:

Air Quality Index (AQI)							
AQI	O ₃ (ppm) 8h	O ₃ (ppm) 1-h ⁽¹⁾	PM _{2.5} (μg/m ³) 24-h	PM ₁₀ (μg/m ³) 24-h	CO (ppm) 8-h	SO ₂ (ppb) 1-h	NO ₂ (ppb) 1-h
Good 0~50	0.000 – 0.054	–	0.0 – 15.4	0–50	0 – 4.4	0–20	0–30
Moderate 51~100	0.055 – 0.070	–	15.5 – 35.4	51–100	4.5 – 9.4	21–75	31–100
Unhealthy for Sensitive Groups 101~150	0.071 – 0.085	0.125 – 0.164	35.5 – 54.4	101–254	9.5 – 12.4	76–185	101–360
Unhealthy 151~200	0.086 – 0.105	0.165 – 0.204	54.5 – 150.4	255–354	12.5 – 15.4	186–304 ⁽³⁾	361–649
Very Unhealthy 201~300	0.106 – 0.200	0.205 – 0.404	150.5 – 250.4	355–424	15.5 – 30.4	305–604 ⁽³⁾	650–1249
Hazardous 301~400	(2)	0.405 – 0.504	250.5 – 350.4	425 – 504	30.5 – 40.4	605–804 ⁽³⁾	1250–1649
Hazardous 401~500	(2)	0.505 – 0.604	350.5 – 500.4	505–604	40.5 – 50.4	805–1004 ⁽³⁾	1650–2049

Figure: Domestic AQI



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Introduction

2 Sliding window

- A technical application tool that can be used for data processing
- Fixing the throughput of the data by the size of the window



Operation

2 Sliding window

The operation will start by inputting a new data from the right side of the window, and remove the oldest data, which is the leftmost one in the window.

Sliding window -->



Slide one element forward



Figure: operation process



Operation

2 Sliding window

This figure shows the moving averages of stock prices with different periods. The orange line is 5 days, the purple line is 10 days, the blue line is 30 days and the pink line is 60 days.



Figure: Moving Average (MA) with different period



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Training and Testing

3 Machine Learning

Consider a dataset of size N , denoted by $\{\mathbf{x}_i, y_i\}_{i=1}^N$. For each $i \in \{1, 2, \dots, N\}$,

$$\mathbf{x}_i = (x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(D)}), x_i^{(j)} \in \mathbb{R}$$

is called a feature vector with D -dimension and $y_i \in \mathbb{R}$ is called a real-valued target.



Linear Regression

3 Machine Learning

In linear regression, we use the model of linear function to predict y .
Consider a linear function given by

$$f_{\mathbf{w},b}(\mathbf{x}) = \mathbf{w}\mathbf{x} + b, \mathbf{w} \in \mathbb{R}^D, y \in \mathbb{R},$$

and solve for $\mathbf{w}^*$ and b^* by minimizing the following equation,

$$\frac{1}{N} \sum_{i=1}^N (f_{\mathbf{w},b}(\mathbf{x}_i) - y_i)^2.$$

Then we can estimate the relationship between $\mathbf{x}$ and y by the linear function.



Linear Regression

3 Machine Learning

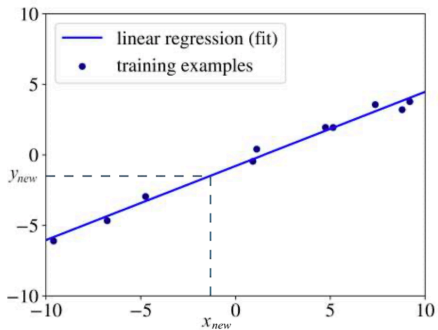


Figure: Linear regression of degree 2

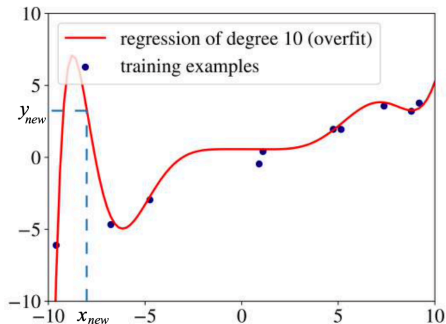


Figure: Linear regression of degree 10



Logistic Regression

3 Machine Learning

In logistic regression, for each $i \in \{1, 2, \dots, N\}$, y_i is either 0 or 1.

For a given $\mathbf{x}$, we desire to predict the probability of y being positive, and define $y = 0$ if y is negative, $y = 1$ otherwise.

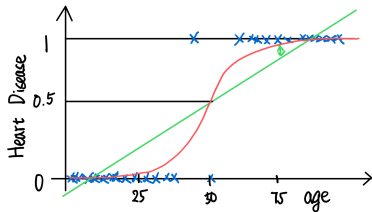


Figure: Example of logistic regression

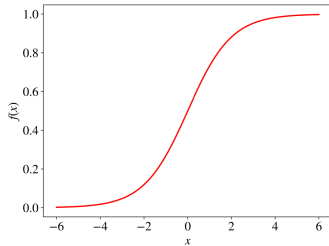


Figure: Graph of $y = \frac{1}{1+e^{-x}}$



Logistic Regression

3 Machine Learning

We consider the model given by

$$f_{\mathbf{w},b}(\mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}\mathbf{x}+b)}}$$

and solve $\hat{\mathbf{w}}$ and $\hat{b}$ by maximize the following likelihood function,

$$L_{\mathbf{w},b} = \prod_{i=1}^N f_{\mathbf{w},b}(\mathbf{x}_i)^{y_i} (1 - f_{\mathbf{w},b}(\mathbf{x}_i))^{(1-y_i)}$$

To calculate this likelihood function, we will like to take nature log on both side, which called log-likelihood function as following:

$$l_{\mathbf{w},b} = \sum_{i=1}^N y_i \ln f_{\mathbf{w},b}(\mathbf{x}_i) + (1 - y_i) \ln(1 - f_{\mathbf{w},b}(\mathbf{x}_i))$$



Decision Tree

3 Machine Learning

A decision tree is a acyclic graph that can be used to make decision. In every branching node of the graph has a specific value k , which examined $\mathbf{x}_i^{(j)}$ for some $j \in \{1, 2, \dots, D\}$, where $i \in \{1, 2, \dots, N\}$. If $\mathbf{x}_i^{(j)} < k$, then the left branch will be followed, otherwise, the right side will be followed.

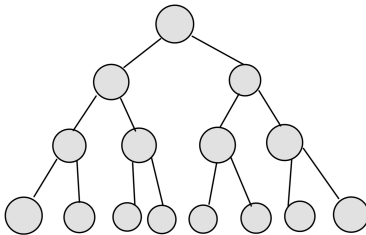


Figure: Decision tree



Decision Tree

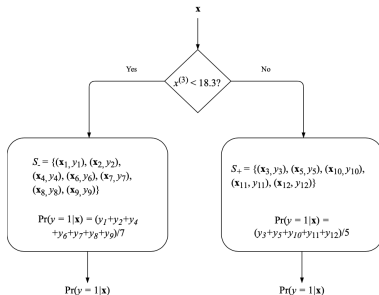
3 Machine Learning

First, we let $f_{ID3} \stackrel{def}{=} Pr(y = 1|\mathbf{x})$ and $S = \{(\mathbf{x}_i, y)\}_{i=1}^N$.

We construct the model:

$$f_{ID3}^S \stackrel{def}{=} \frac{1}{|S|} \sum_{(\mathbf{x}, y) \in S} y$$

which have the same output for any $\mathbf{x}$ in S .





Decision Tree

3 Machine Learning

Then we want to split S into two sets $S_- = \{(\mathbf{x}, y) | (\mathbf{x}, y) \in S, x^{(j)} < k\}$ and $S_+ = \{(\mathbf{x}, y) | (\mathbf{x}, y) \in S, x^{(j)} \geq k\}$,

which is the best split evaluated by the entropy function as following:

$$H(S) = -f_{ID3}^S \ln f_{ID3}^S - (1 - f_{ID3}^S) \ln(1 - f_{ID3}^S)$$

which attains it's maximum when S is equiprobable (有一半的例子 $y = 1$), attains it's minimum when there only have one value ($y = 1$ for all $(\mathbf{x}, y)$, or $y = 0$ for all $(\mathbf{x}, y)$)

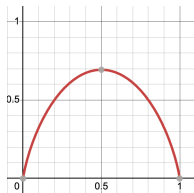


Figure: Entropy function



Decision Tree

3 Machine Learning

When we split a example set by some j, k , we have a entropy function of the split define as following

$$H(S_-, S_+) = \frac{|S_-|}{|S|} H(S_-) + \frac{|S_+|}{|S|} H(S_+). \quad (1)$$

In conclusion, when ever we decide a node of decision tree, we find the split by minimizing equation 1.

Stopping Condition:

1. Every leaf node have the same value y .
2. Cannot find a attribute to split the example.
3. The split entropy less then a given ε .
4. The tree reaches the maximum depth given by d .



Support Vector Machine

3 Machine Learning

In SVM, for each $i \in \{1, 2, \dots, N\}$, y_i is either -1 or 1 .

The hyperplane h is given by $\mathbf{w} \in \mathbb{R}^D$, $b \in \mathbb{R}$ as following:

$$h : \mathbf{w}^T \mathbf{x} + b = 0$$

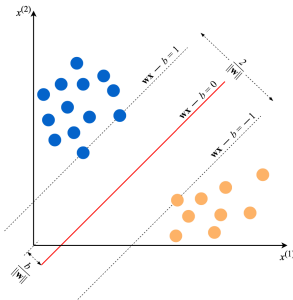


Figure: SVM (Hard margin SVM)



Support Vector Machine

3 Machine Learning

There are two questions can be ask.

1. What if there are noises such that there is no hyperplane that can separate the positive ones perfectly and the negative ones?
2. What if the dataset can not be separate by a plane, but can be separate perfectly by a high-order polynomial?

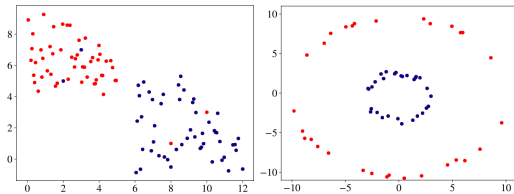


Figure: Two question two deal with



Dealing with noise

3 Machine Learning

For the first question, we will introduce soft margin SVM.

Before showing the equation of soft margin SVM, we first introduce the hinge loss function as following:

$$\max_{i=1,2,\dots,N} (0, 1 - y_i(\mathbf{w}\mathbf{x}_i + b))$$

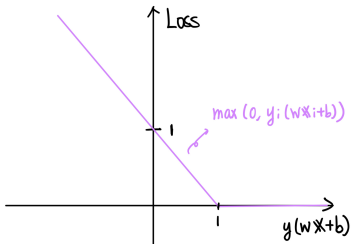


Figure: Hinge loss function

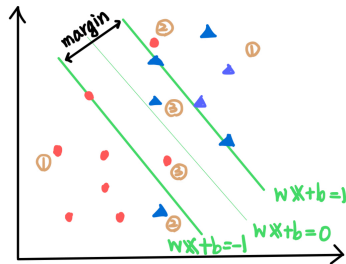


Figure: Soft Margin SVM



Dealing with noise

3 Machine Learning

After knowing the hinge loss function, we can come up the formula for soft margin SVM is to minimize the function as following:

$$\lambda \|w\|^2 + \frac{1}{N} \sum_{i=1}^N \max_{i=1,2,\dots,N} (0, 1 - y_i(\mathbf{w}\mathbf{x}_i + b))$$

The λ in front of $\|w\|^2$ is a positive value such that decide rather we want a bigger margin or less noise. If λ is large, then the term of loss function will negligible. And the choose of λ is base on experience as the ϵ and d in the stopping condition of decision tree.



Dealing with Inherent Non-Linearity

3 Machine Learning

We first take a look at the example of a dataset which can not be separated by a line (hard margin SVM) as following:

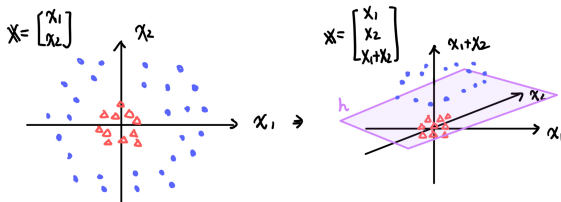


Figure: Example of inherent non-linear



Dealing with Inherent Non-Linearity

3 Machine Learning

By achieving this goal, we first need to do a little change on the formula of getting hard margin SVM, which is

$$\min \|\mathbf{w}\|^2$$

such that

$$\min_{i=1,2,\dots,N} y_i(\mathbf{x}_i + b) = 1.$$

By using the method of Lagrange multiplier,

$$L(\mathbf{w}, b, \alpha) = \frac{1}{2} \mathbf{w}^T \mathbf{w} - \sum_{i=1}^N \alpha_i (y_i(\mathbf{w} \mathbf{x}_i + b) - 1)$$

, where α_i is called Lagrange multiplier.



Dealing with Inherent Non-Linearity

3 Machine Learning

We can get a equivalent function as hard margin SVM as following:

$$\max_{\alpha_i \geq 0} [\min_{\mathbf{w}, b} L(\mathbf{w}, b, \alpha_i)] \quad (2)$$

By calculating the min inside the brackets, we can get two equation:

$$\frac{\partial L}{\partial \mathbf{w}} = 0 \implies \mathbf{w} = \sum_{i=1}^N \alpha_i y_i \mathbf{x}_i$$

$$\frac{\partial L}{\partial b} = 0 \implies \sum_{i=1}^N \alpha_i y_i = 0 = \alpha^T \mathbf{y}$$

Note that since it is hard margin in this case, so the determine of $\mathbf{w}$ only depends on the support vectors, which are the vectors such that $y_i(\mathbf{w}\mathbf{x}_i + b) = 1$.



Dealing with Inherent Non-Linearity

3 Machine Learning

We keep on calculating equation 2 by plugging the method of Lagrange multiplier, we can get

$$\max_{\alpha_i \geq 0} [\min_{\mathbf{w}, b} L(\mathbf{w}, b, \alpha_i)] = \max_{\alpha_i \geq 0} [\sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j] = \max_{\alpha_i \geq 0} [\sum_{SV} \alpha_i - \frac{1}{2} \sum_{SV} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^T \mathbf{x}_j]$$

We can know the only information we need to know is about the pair inner product of any two support vectors. Which is called SVM dual.

The reason using SVM dual rather than hard margin SVM is because in real world, we may have not many examples(N), however every example has many character(D), such that $D \gg N$.



Dealing with Inherent Non-Linearity

3 Machine Learning

So, as we show in Figure 13, we want to find a transform

$T[x^{(1)}, x^{(2)}]^T = [1, x^{(1)}, x^{(2)}, x^{(1)}x^{(2)}, \dots]^T$ such that we can find the hyperplane of the dataset.

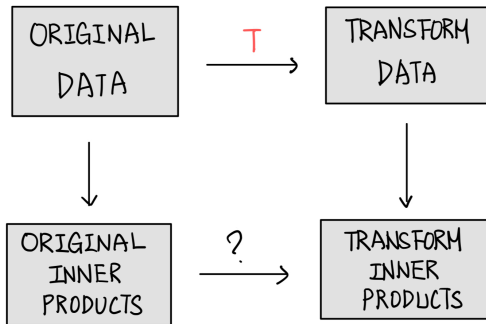


Figure: Process of Kernels



Dealing with Inherent Non-Linearity

3 Machine Learning

The solution is consider a function define as following:

$$K_{poly(n)}(\mathbf{x}_i \cdot \mathbf{x}_j) = (1 + \mathbf{x}_i \cdot \mathbf{x}_j)^n = \left(1 + \sum_{k=1}^D x_i^{(k)} x_j^{(k)}\right)^n \quad (3)$$

We can find out that the equation 3, which is called Kernel SVM, gives the same terms as we do the transform first and calculate the inner product.

Finally, base on above, we can define the Radial Basis Function Kernel

$$\begin{aligned} K_{RBF}(\mathbf{x}_i \cdot \mathbf{x}_j) &= e^{-\frac{1}{2} \|\mathbf{x}_i - \mathbf{x}_j\|^2} \\ &= e^{-\frac{1}{2} (\mathbf{x}_i^T \mathbf{x}_i + \mathbf{x}_j^T \mathbf{x}_j)} e^{\mathbf{x}_i^T \mathbf{x}_j} \\ &= C' e^{(1 + \mathbf{x}_i^T \mathbf{x}_j)} \\ &= C' \sum_{n=1}^{\infty} \frac{(1 + \mathbf{x}_i^T \mathbf{x}_j)^n}{n!} = C' \sum_{n=1}^{\infty} \frac{K_{poly(n)}(\mathbf{x}_i \cdot \mathbf{x}_j)}{n!} \end{aligned} \quad (4)$$



k-Nearest Neighbors

3 Machine Learning

k -Nearest Neighbors (kNN) unlike other learning algorithm that can discard the training data after the model was made. kNN keeps all the training data, when a previously unseen example $\mathbf{x}$ was input, kNN will find k training examples which is closest to $\mathbf{x}$, and return majority label or average label to $\mathbf{x}$'s label γ .



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Data

4 Reason for Data Collection

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	date_name	降水量	風速(m/s)	風向	氣溫	相對溼度(%)	紫外線	AQI	SO2	CO	PM10	NO2	O3(8hr)	PM2.5
2	2023/10/31	0	2.4	58	24.3	69	8	39	2	2	11	22	39	20
3	2023/10/30	0	1.4	52	23.5	74	6	38	5	3	14	33	38	26
4	2023/10/29	0	1.4	56	22.9	70	5	40	5	3	16	23	40	29
5	2023/10/28	14.5	1.8	53	21.9	78	4	38	2	3	10	22	38	17
6	2023/10/27	0	1.8	50	24.3	72	5	38	2	3	18	32	38	32
7	2023/10/26	0	1.5	92	25.1	68	8	80	5	5	23	38	80	49
8	2023/10/25	0	1.9	38	25.6	75	8	38						
9	2023/10/24	0	1.3	59	25.6	77	6	50	5	5	24	35	50	47
10	2023/10/23	0	1.2	87	25.8	71	8	46	5	3	14	38	46	24
11	2023/10/22	0	1.8	50	23.9	71	6	34	2	2	7	18	34	13
12	2023/10/21	0	2.9	57	22.5	72	5	37	2	3	14	18	37	22
13	2023/10/20	0	2	1	23.9	85	5	51	2	3	23	25	40	51
14	2023/10/19	0	0.9	21	25.9	76	7	169	5	6	44	28	169	94
15	2023/10/18	0	0.8	20	25.8	70	8	140	8	7	39	60	140	81
16	2023/10/17	0	1.8	32	25.4	56	8	126	2	5	32	22	126	67
17	2023/10/16	0	2.3	55	24.9	55	10	90	2	3	32	17	90	65
18	2023/10/15	0	2	58	24.1	62	8	108	5	5	30	22	108	65
19	2023/10/14	0	1.5	52	24.2	70	5	64	8	5	22	28	64	45
20	2023/10/13	0	1.8	29	25.2	68	9	54	2	3	22	28	54	29
21	2023/10/12	0	2.2	53	25.1	67	9	47	5	3	20	25	47	29
22	2023/10/11	0.5	3	58	24.6	67	10	58	2	3	18	18	58	27
23	2023/10/10	0	2.4	30	24.6	81	11	40	2	2	8	15	40	13
24	2023/10/9	0	2	53	24.5	87	3	42	0	2	12	13	42	18



Confusion matrix

4 Reason for Data Collection

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

	Actual positive	Actual negative
Predicted positive	TP	FP
Predicted negative	FN	TN



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Result of Machine Learning

5 Result of Machine Learning



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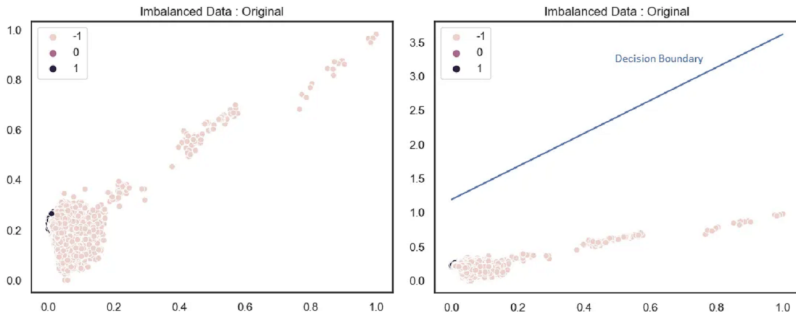
6 Improving data

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Imbalanced Data

6 Improving data





F-score

6 Improving data

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

With these two indicators, we can define F1-Score or use F_{β} - Score to adjust the ratio of precision and recall.

$$F1 - \text{Score} = \frac{2 * P * R}{P + R}$$

$$F_{\beta} - \text{Score} = \frac{(1 + \beta^2) * PR}{P * \beta^2 + R}$$

where β is the ratio of precision and recall.



Precision-Recall Curves

6 Improving data

PR curve, plots recall on the x-axis and precision on the y-axis. And define AUPRC to evaluate the model's performance by calculating the area under the PR curve.

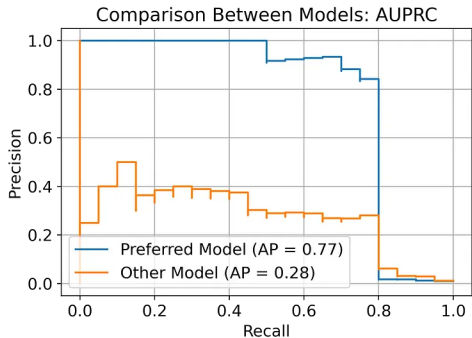


Figure: PR curve and AUPRC



ROC Curves

6 Improving data

$$FPR = \frac{FP}{TN + FP}, TPR = \frac{TP}{TP + FN}$$

ROC curve, plots FPR on the x-axis and TPR on the y-axis. And define AUC to evaluate the model's performance by calculating the area under the ROC curve.

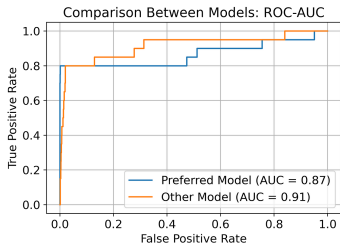


Figure: ROC curve and AUC



Oversampling and Undersampling

6 Improving data

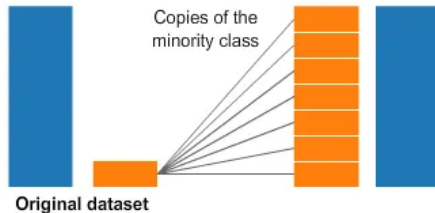
Oversampling: generating synthetic data points for the minority class.

Undersampling: reducing the number of instances in the majority class. In reality, we can combine two types of method to sample data.

Undersampling



Oversampling

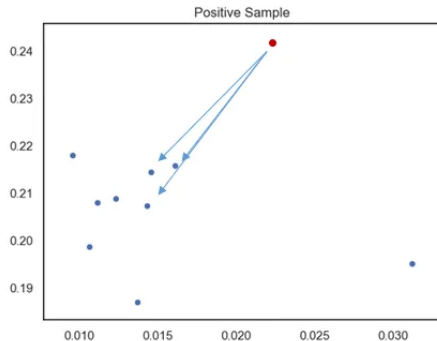




Oversampling: SMOTE

6 Improving data

1. Pick up one point of the minority class, call $x_{nearest}$.
2. Define the number of neighbors, k . This means that we need to find the k nearest points to $x_{nearest}$.



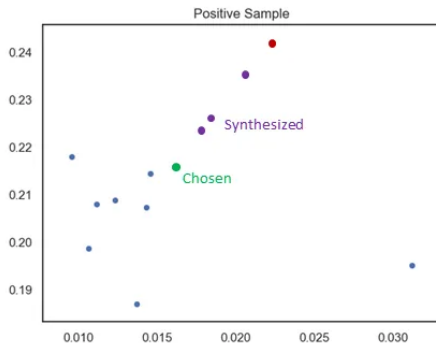


Oversampling: SMOTE

6 Improving data

3. Randomly choose one point of the k nearest points, called x_{chosen} .
4. Generate new sample point x_{new} by

$$x_{new} = x_{chosen} + (x_{nearest} - x_{chosen}) \cdot \delta; \delta \in [0, 1]$$

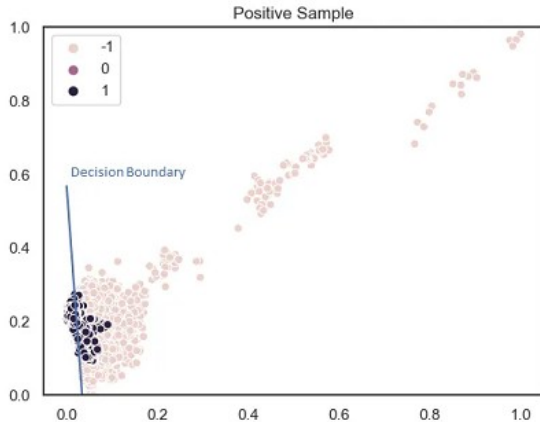
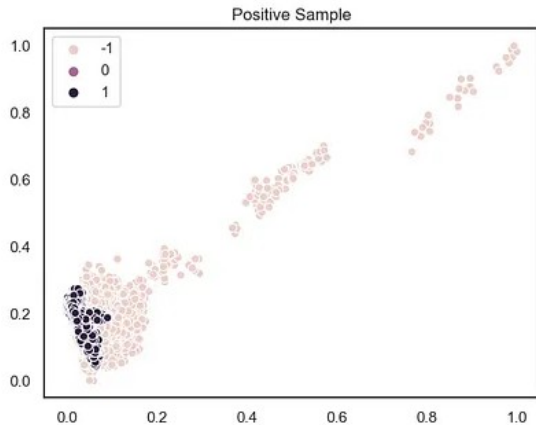




Oversampling: SMOTE

6 Improving data

5. Apply to all points of the minority class.





Undersampling: Tomek Link

6 Improving data

For an instance pair (x_i, x_j) where $x_i \in S_{max}$ and $x_j \in S_{min}$, if there is no x_k such that $d(x_i, x_k) < d(x_i, x_j)$ or $d(x_j, x_k) < d(x_i, x_j)$, then (x_i, x_j) is a tomek link. After identifying all tomek links, the corresponding x_i in each link is then removed.

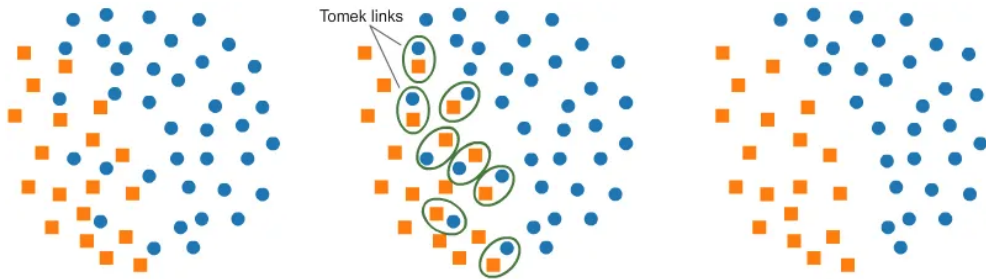


Figure: how to find tomek link



Application

6 Improving data

We use the extension of RapidMiner Studio, which includes SMOTE method and performance of AUC and AUPRC, to predict AQI again.

accuracy: 76.23%

	true false	true true	class precision
pred. false	68	34	66.67%
pred. true	29	134	82.21%
class recall	70.10%	79.76%	

AUC: 0.810、AUPRC: 0.843

Figure: 經過 SMOTE 處理、使用前一日 AQI 預測 AQI



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LightGBM

7 Literature Review

LightGBM uses tree-based learning strategies, making it a suitable choice for gradient boosting. It provides quicker training and higher production thanks to its decentralised and effective architecture.

When being trained, it works more quickly and can handle large and complex datasets. Support for both parallel and GPU-based learning. The method enables one to infer details about a target Y given only X as input when used in supervised learning scenarios. The LightGBM technique uses a supervised training set (X) and a loss function $L(y, f(x))$ whose predicted value is to be reduced to accomplish this $\hat{f}(x)$.

$$\hat{f} = \arg \min_f E_{y,x} L(y, f(x))$$



Random Forest

7 Literature Review

Random forest regression is one form of machine learning-based regression technique. The foundation of this tactic is a combination of bagging and random subspace approaches. Following the bagging of numerous learning trees, an ensemble method is used to merge these trees into a single forecast. From the entire collection of N examples used for training (D), n random instances are chosen to create a bootstrap sample (D_b). It is acceptable to substitute examples when creating bootstrap samples. The input vector x is used to create K distinct regression trees for the bootstrap samples. When performing regression tasks, the random forest prediction is made by averaging K predictions from regression trees $h_k(x)$.



CatBoost

7 Literature Review

- Boosting is a method based on the idea that multiple relatively weak models can be combined to create a single, highly competitive predictive model that performs better than random guessing.
- Gradient boosting reduces errors by fitting a series of decision trees, each learning from the errors of the previous iteration. This process of introducing new functions is repeated until the selected loss function can no longer be minimized.
- CatBoost differs from traditional gradient boosting models in the way it creates decision trees. In contrast, CatBoost creates "ignorant trees" because all nodes in the same layer are required to test the same predictor under the same conditions. This allows the index of the leaf to be calculated using only bit operations.



Adaptive Boosting (AdaBoost) Regressor

7 Literature Review

AdaBoost modifies the data of each training sample (x_i, y_i) by applying a weight $w_1, w_2, \dots, w_N$.

In this method, the predictions made by the base learner after the weak learner are more likely to be accurate. This process will be carried out again until the t th iteration, after which the T base learning algorithm's limit will be reached. The outputs of weak learners can be combined to generate more robust learners, which enhance the ability to predict outcomes.



Extreme Gradient Boosting (XGBoost)

7 Literature Review

Gradient Boosting's XGBoost variation outperforms the original in terms of computing efficiency, scalability, and generalisation performance.

All category data will be transformed into their numeric equivalents, because XGBoost only takes numeric vectors as input. This encoding change can be made with a single hot encoding. The feature engineering and data purification stages come next. We can obtain the estimated model by using the universal function, as indicated by the following formula.



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International AQI

8 International AQI

The International Air Quality Index:

The Air Quality Index (AQI) is calculated based on monitoring data for various pollutants in the air on a given day.

The AQI for a specific monitoring station on a given day is determined by the highest sub-index value among these pollutants. The calculation of AQI among different countries is essentially same and based on this formula:

$$IAQI_P = \frac{IAQI_{Hi} - IAQI_{Lo}}{BP_{Hi} - BP_{Lo}}(C_P - BP_{Lo}) + IAQI_{Lo}$$



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Conclusion

9 Conclusion

By researching the issue, we first gained a deeper understanding of air pollution and learned many machine learning methods and algorithms. We used one of the methods - SVM to use relevant data to predict AQI values. However, from the operation results it can be seen from the picture that when SVM uses a single factor to predict AQI, the accuracy is only about seventy percent. After discussion, we came to a possible reason: SVM cannot distinguish between right and wrong due to the data, so it guesses all right.



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Numerical Analysis Project

Thank you for listening!
Any questions?